

On the large-clique version of the Erdős-Sós conjecture^{*}

Kun Cheng Yurui Tang[†]

Department of Mathematics, East China Normal University, Shanghai, 200241, China

Abstract

For graphs H and F , let $\text{ex}(n, H, F)$ denote the maximum number of copies of H in an F -free graph of order n . Motivated by the Erdős-Sós conjecture, Gerbner and Palmer and, independently, Zhao and Peng conjectured that for every tree T of order k and every $3 \leq r \leq k-1$,

$$\text{ex}(n, K_r, T) = a \binom{k-1}{r} + \binom{b}{r},$$

where $n = a(k-1) + b$ with $0 \leq b < k-1$. In this paper, we confirm the conjecture for $r \geq \lceil (2k-1)/3 \rceil$ and characterize all extremal graphs.

Keywords: Generalized Turán problem; Tree; Erdős-Sós conjecture; Clique counting

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1. Introduction

We consider finite simple graphs and use standard terminology and notation from [2], [3] and [24]. Let G be a graph with vertex set $V(G)$ and edge set $E(G)$. Then $|G| := |V(G)|$ is called the *order* of G . For $v \in V(G)$, let $N_G(v)$ be the neighborhood of v in G . We write K_n for the complete graph of order n . Two graphs are *disjoint* if they have no vertex in common. The *union* of simple graphs G and H is the graph $G \cup H$ with vertex set $V(G) \cup V(H)$ and edge set $E(G) \cup E(H)$. If G and H are disjoint, we refer to their union as a *disjoint union*, and generally denote it by $G + H$. For graphs we will use equality up to isomorphism, so $G = H$ means that G and H are isomorphic. A *tree* is a connected graph without cycles, and a *forest* is a graph without cycles.

For graphs H and F , let $\mathcal{N}(H, G)$ denote the number of copies of H in a graph G , and let

$$\text{ex}(n, H, F) = \max\{\mathcal{N}(H, G) : |G| = n \text{ and } G \text{ is } F\text{-free}\}.$$

An n -vertex F -free graph attaining this maximum is called an *extremal graph*. Thus the classical Turán number $\text{ex}(n, F)$ is the case $H = K_2$. For each non-bipartite graph F , the celebrated Erdős-Stone-Simonovits theorem [9, 10] determines an asymptotic formula for $\text{ex}(n, F)$. Hence it is natural and interesting to study the Turán number $\text{ex}(n, B)$ of a bipartite graph B . For more advances in this topic, we refer the reader to the survey [13]. In particular, let T be a tree of order k , considering the disjoint copies of K_{k-1} shows that $\text{ex}(n, T) \geq \frac{n(k-2)}{2}$. The following is a well-known conjecture.

Conjecture 1.1 (Erdős-Sós [8]). *Let T be a tree of order k . The bound*

$$\text{ex}(n, T) \geq \frac{n(k-2)}{2}$$

is tight.

^{*}Email addresses: chengkunmath@163.com (K. Cheng), tyr2290@163.com (Y. Tang).

[†]Corresponding author

Conjecture 1.1 is still open, however it has been confirmed in some special cases. For example, Conjecture 1.1 was shown to be true when G is C_4 -free [20], P_{k+5} -free [7]; when k is large compared to n [17, 21, 22, 25, 27, 30]; for some special classes of trees [11, 12, 16, 19], and for spectral version [4, 6].

The systematic study of this generalized Turán function was initiated by Alon and Shikhelman [1]; see papers [5, 14, 18, 23, 26, 28, 32] and a nice survey [15].

Let T be a tree of order k , and let $3 \leq r \leq k-1$. If $n = a(k-1) + b$, where $0 \leq b < k-1$, then the graph $aK_{k-1} + K_b$ is T -free, and

$$\mathcal{N}(K_r, aK_{k-1} + K_b) = a \binom{k-1}{r} + \binom{b}{r}.$$

Gerbner and Palmer [15, Conjecture 6.5] and, independently, Zhao and Peng [29] conjectured that this construction is always extremal.

Conjecture 1.2 (Gerbner-Palmer [15], Zhao-Peng [29]). *Let T be a tree of order k , let $3 \leq r \leq k-1$ and let $n = a(k-1) + b$ with $0 \leq b < k-1$. Then*

$$\text{ex}(n, K_r, T) = a \binom{k-1}{r} + \binom{b}{r}.$$

Zhou and Yuan [31] verified Conjecture 1.2 for $r = k-2$ and for $r = k-3 \geq 5$, and in those cases they also characterized the extremal graphs. Zhao and Peng [29] confirmed Conjecture 1.2 for $r = k-d$, where $d \geq 2$ and $k \geq d^2 - d + 3$, and characterized all extremal graphs. In this paper, we show that Conjecture 1.2 holds for a range whose width grows linearly with k , and characterize all extremal graphs. Our main result is as follows.

Theorem 1.1. *Conjecture 1.2 holds for $r \geq \lceil (2k-1)/3 \rceil$. Moreover, the extremal graphs are exactly the following:*

- (i) *If $0 \leq b < r$, then $G = aK_{k-1} + H$, where H is an arbitrary graph of order b ;*
- (ii) *If $r \leq b < k-1$, then $G = aK_{k-1} + K_b$.*

2. Preliminaries

A graph G is called a *split graph* if there exists a partition (V_1, V_2) of $V(G)$ such that V_1 is a clique and V_2 is an independent set. In addition, if every vertex of V_1 is adjacent to every vertex of V_2 , then G is called a *complete split graph*.

We first pose a lemma about tree partition.

Lemma 2.1. *Let k and r be two positive integers with $r \geq \lceil (2k-1)/3 \rceil$, and let T be a tree of order k . Then T contains a vertex u such that the components of $T - u$ can be divided into two families that respectively contain at most $r-1$ vertices.*

Proof. If $r \geq k$, the result holds trivially. We next consider the case $r \leq k-1$. We show that there exists a vertex u of T such that each component of $T - u$ has order at most $k/2$. For each $w \in V(T)$, let

$$m(w) = \max\{|T'| : T' \text{ is a component of } T - w\}.$$

Choose $u \in V(T)$ such that $m(u)$ is minimum. Suppose to the contrary that $m(u) > k/2$. Let T^* be the component of $T - u$ with $|T^*| = m(u)$. Let v be the unique neighbor of u in T^* . In $T - v$, the component containing u has order $k - m(u) < m(u)$, while every other component is contained in $T^* - v$ and hence has order at most $m(u) - 1$. Thus $m(v) < m(u)$, which contradicts the choice of u . Therefore, $m(u) \leq k/2$.

Next we are to show that the components of $T - u$ can be divided into two families that respectively contain at most $r - 1$ vertices. Let $C_1, \dots, C_m$ be the components of $T - u$ with $a_i = |C_i|$, where $a_1 \leq \dots \leq a_m$. Then

$$a_i \leq \frac{k}{2} \leq r - 1 \text{ for each } 1 \leq i \leq m, \text{ and } \sum_{i=1}^m a_i = k - 1. \quad (2.1)$$

If $a_m \geq k - r$, let the first family consist of that component C_m and let the second family consist of all remaining components. By (2.1), the first family has order at most $r - 1$, whereas the second has order at most $k - 1 - (k - r) = r - 1$.

We may therefore assume that $a_m \leq k - r - 1$. Hence $a_i \leq k - r - 1$ for each $1 \leq i \leq m$. Add components to the first family until their total order is at least $k - r$. Since $a_i \leq k - r - 1$ and $\lceil (2k - 1)/3 \rceil \leq r$, it follows that the first family has order at most $2(k - r - 1) \leq r - 1$. Similarly, the second family consists of all remaining components and hence has order at most $k - 1 - (k - r) = r - 1$. This proves Lemma 2.1. $\square$

Using Lemma 2.1, we get the following.

Proposition 2.2. *Let r, k be two positive integers with $r \geq \lceil (2k - 1)/3 \rceil$, and let T be a tree of order k . Let G_1 and G_2 be two complete graphs of order r . If $V(G_1) \cap V(G_2) \neq \emptyset$ and $|V(G_1) \cup V(G_2)| \geq k$, then $G_1 \cup G_2$ contains the tree T .*

Proof. For the tree T , by Lemma 2.1, there exists a vertex $u \in V(T)$ such that $T - u$ contains two families of components with order s and $k - 1 - s$, respectively, where $\max\{s, k - 1 - s\} \leq r - 1$. Since $V(G_1) \cap V(G_2) \neq \emptyset$, there exists a vertex $x \in V(G_1) \cap V(G_2)$ corresponding to the vertex $u \in V(T)$.

If $|V(G_1) \setminus V(G_2)| \geq s$, then $G_1 - V(G_2)$ contains the subgraph of T induced by the first family, and $G_2 - x$ contains the subgraph of T induced by the second family, because $|G_2 - x| = r - 1$. Thus $G_1 \cup G_2$ contains T .

If $|V(G_1) \setminus V(G_2)| < s$, then we may choose a vertex subset U with $V(G_1) \setminus V(G_2) \subseteq U \subseteq V(G_1)$ and $|U| = s$ such that the subgraph of G_1 induced by U contains the subgraph of T induced by the first family. Then $(V(G_1) \cup V(G_2)) \setminus (\{x\} \cup U) = V(G_2) \setminus (\{x\} \cup U)$ has order at least $k - 1 - s$, and thus contains the subgraph of T induced by the second family. This proves Proposition 2.2. $\square$

Lemma 2.3. *Let $c \geq 1$ be an integer and let F be a forest of order at least $2c$. Then F contains an independent set S such that $|S| = c$ and $|N_F(S)| \leq c$.*

Proof. We use induction on c . If $c = 1$, the assertion is trivial. Assume $c \geq 2$ and the result holds for $c - 1$. If F has an isolated vertex u , then $F - u$ has at least $2(c - 1)$ vertices. By the induction hypothesis, $F - u$ contains an independent set S' such that $|S'| = c - 1$ and $|N_{F-u}(S')| \leq c - 1$. Set $S = S' \cup \{u\}$. Then $|S| = c$ and $|N_F(S)| \leq c$.

We next assume that F has no isolated vertices. Choose a leaf $u \in F$ and let v be its unique neighbour. The forest $F' = F - \{u, v\}$ has at least $2(c - 1)$ vertices. By the induction hypothesis, F' has an independent set S' of order $c - 1$ satisfying $|N_{F'}(S')| \leq c - 1$. Since $N_F(u) = \{v\}$, the set $S = S' \cup \{u\}$ is independent and $N_F(S) \subseteq N_{F'}(S') \cup \{v\}$. Hence $|N_F(S)| \leq c$. This proves Lemma 2.3. $\square$

By Lemma 2.3, we obtain the following.

Proposition 2.4. *Let k, c be two positive integers with $k \geq 2c$, and let F be a forest of order k . Let G_1 be a complete graph of order at least $k - c$, and let G_2 be a complete split graph of order $2c$ with $V(G_2) = R \cup Y$, where R is a c -clique and Y is an independent set of size c . If $V(G_1) \cap V(G_2) = R$, then $G_1 \cup G_2$ contains the forest F .*

Proof. From Lemma 2.3, there exists an independent set S in the forest F such that $|S| = c$ and $|N_F(S)| \leq c$. Then the subgraph of $G_1 \cup G_2$ induced by Y contains the subgraph of F induced by S . Since $|N_F(S)| \leq c = |R|$, then the subgraph of $G_1 \cup G_2$ induced by $V(G_1)$ contains the subgraph of F induced by $V(F) \setminus S$. Note that G_1 is a complete graph and G_2 is a complete split graph. Therefore, $G_1 \cup G_2$ contains the forest F . This completes the proof of Proposition 2.4. $\square$

The following lemma will be needed.

Lemma 2.5 (Zhao-Peng [29], Zhou-Yuan [31]). *Let $k - 1 \geq r \geq 2$, let $a \geq 0$, and let $0 \leq b < k - 1$. Assume that $x_1, \dots, x_m$ are integers with $0 \leq x_i \leq k - 1$ and $\sum_{i=1}^m x_i \leq a(k - 1) + b$. Then*

$$\sum_{i=1}^m \binom{x_i}{r} \leq a \binom{k-1}{r} + \binom{b}{r}.$$

Equality holds only if the following conditions hold:

- (i) *if $b < r$, then $m = a$ and $x_1 = \dots = x_a = k - 1$;*
- (ii) *if $b \geq r$, then the multiset $\{x_1, \dots, x_m\} = \underbrace{\{k - 1, \dots, k - 1\}}_a, b\}$.*

The following lemma is an immediate consequence of Vandermonde's identity.

Lemma 2.6. *For nonnegative integers m, n and s with $\min\{m, n\} \geq s$,*

$$\sum_{j=0}^s \binom{m}{j} \binom{n}{s-j} = \binom{m+n}{s}.$$

3. Proofs

The following notations are frequently used in the forthcoming discussions. For a graph G , let $\mathcal{K}_r(G)$ denote the family of r -cliques of G . Define an auxiliary graph $\Gamma_r(G)$ with vertex set $\mathcal{K}_r(G)$, where two members are adjacent if they intersect. Then $V(\mathcal{C})$ is the vertex set of a component $\mathcal{C}$ of $\Gamma_r(G)$. Hence, $V(\mathcal{C}) \subseteq \mathcal{K}_r(G)$. The components of $\Gamma_r(G)$ are called the K_r -components of G . For a K_r -component $\mathcal{C}$ of G , let $U(\mathcal{C}) = \bigcup_{A \in V(\mathcal{C})} A$. Hence, $U(\mathcal{C}) \subseteq V(G)$, and the sets $U(\mathcal{C})$ corresponding to distinct K_r -components are vertex-disjoint.

Lemma 3.1. *Let r and k be two integers with $k - 1 \geq r \geq \lceil (2k - 1)/3 \rceil$. Let T be a tree of order k , and let G be a T -free graph.*

- (i) *For any pair r -cliques A and B in G , if $A \cap B \neq \emptyset$, then $|A \cup B| \leq k - 1$. Moreover, $|A \cap B| \geq 2r - k + 1$.*
- (ii) *Any two members of the same K_r -component of G intersect at least $2r - k + 1$ vertices.*

Proof. (i) Note that $|A| = |B| = r$ and $|A \cap B| \geq 1$. If $|A \cup B| \geq k$, then by Proposition 2.2, $G[A \cup B]$ contains a subgraph isomorphic to T , a contradiction. Therefore $|A \cup B| \leq k - 1$, and hence $|A \cap B| = |A| + |B| - |A \cup B| \geq 2r - k + 1$.

(ii) Note that a graph is connected if and only if there is a path between any two vertices. Let $A_0 A_1 \dots A_t$ be a path in a K_r -component, where $A_0, A_1, \dots, A_t$ are r -cliques of G . It suffices to prove that $|A_i \cap A_j| \geq 2r - k + 1$ for $i, j \in \{0, 1, 2, \dots, t\}$.

We assert that $|A_0 \cap A_i| \geq 2r - k + 1$. Use induction on i . If $i = 1$, then by (i), $|A_0 \cap A_1| \geq 2r - k + 1$. Suppose it holds for $i - 1$. Thus, $|A_0 \cap A_{i-1}| \geq 2r - k + 1$. By (i), $|A_{i-1} \cap A_i| \geq 2r - k + 1$. Hence

$$|A_0 \cap A_i| \geq 2(2r - k + 1) - r = 3r - 2k + 2 \geq 1,$$

where the last inequality is exactly $r \geq \lceil (2k - 1)/3 \rceil$. Thus A_0 and A_i intersect. By (i), it follows that $|A_0 \cap A_i| \geq 2r - k + 1$. $\square$

We are now ready to prove Theorem 1.1.

Proof of Theorem 1.1. Let T be a tree of order k . The graph $aK_{k-1} + K_b$ is T -free and hence

$$\text{ex}(n, K_r, T) \geq \mathcal{N}(K_r, aK_{k-1} + K_b) = a \binom{k-1}{r} + \binom{b}{r}.$$

Let G be an arbitrary T -free graph of order n . We are to show that $\mathcal{N}(K_r, G) \leq a \binom{k-1}{r} + \binom{b}{r}$. Let $\mathcal{C}_1, \dots, \mathcal{C}_m$ be all components of $\Gamma_r(G)$, that is, $\mathcal{C}_1, \dots, \mathcal{C}_m$ are all K_r -components of G .

Claim 1. $|\mathcal{C}_i| \leq \binom{k-1}{r}$ for $1 \leq i \leq m$, with equality if and only if $G[U(\mathcal{C}_i)] = K_{k-1}$.

Proof of Claim 1. Choose an arbitrary K_r -component $\mathcal{C}$ of G . Choose an r -clique $A \in \mathcal{C}$. Let $X = U(\mathcal{C}) \setminus A$ and let $t = 2r - k + 1$. Recall that $r \leq k - 1$. Then $t \leq r$. Let $c = k - r$. Since $r \geq \lceil (2k - 1)/3 \rceil$, it follows that $2r - k + 1 \geq k - r$, i.e., $t \geq c$. For any subset $Q \subseteq A$, write $\tilde{N}_X(Q) = X \cap \bigcap_{v \in Q} N_G(v)$. We assert that

$$|\tilde{N}_X(Q)| \leq c - 1, \text{ for each } Q \subseteq A \text{ with } |Q| \geq t. \quad (3.2)$$

Suppose to the contrary that $|\tilde{N}_X(Q)| \geq c$ for some $Q \subseteq A$ with $|Q| \geq t$. Choose a set $Y \subseteq \tilde{N}_X(Q)$ with $|Y| = c$. Since $c \leq t$, we can choose a subset $R \subseteq Q$ with $|R| = c$. Then R is a clique. Since $Y \subseteq \tilde{N}_X(Q)$, each vertex of Y is adjacent to R . Note $|A| = r = k - c$. Since $r \geq \lceil (2k - 1)/3 \rceil$, we have $k \geq 2(k - r) = 2c$. Then by Proposition 2.4, $G[A \cup Y]$ contains the tree T , a contradiction. Thus, (3.2) holds.

For $0 \leq j \leq c - 1$, write $\mathcal{B}_j := \{B \in V(\mathcal{C}) : |B \setminus A| = j\}$. Next we determine $|\mathcal{B}_j|$. By Lemma 3.1(ii), every r -clique $B \in V(\mathcal{C})$ satisfies $|A \cap B| \geq t$. Choose a $B \in \mathcal{B}_j$ and let $Q := A \cap B$. Then $r - j = |Q| \geq t$ and $B \setminus A \subseteq \tilde{N}_X(Q)$. By (3.2), $|\tilde{N}_X(Q)| \leq c - 1$. For each fixed $Q \subseteq A$, there are at most $\binom{c-1}{j}$ choices for $B \setminus A$, while there are $\binom{r}{r-j} = \binom{r}{j}$ choices for Q . Hence

$$|\mathcal{B}_j| \leq \binom{r}{j} \binom{c-1}{j} \text{ for } 0 \leq j \leq c - 1. \quad (3.3)$$

Summing over j and applying Lemma 2.6 gives

$$|\mathcal{C}| \leq \sum_{j=0}^{c-1} \binom{r}{j} \binom{c-1}{j} = \binom{r+c-1}{c-1} = \binom{k-1}{k-1-r} = \binom{k-1}{r},$$

which proves the first part of Claim 1.

Next we consider the equality case in Claim 1. Then every inequality above becomes equality. The sufficiency is immediate. We now prove the necessity.

If $c = 1$, then by (3.2), $V(\mathcal{C}) = \{A\}$. Thus $G[U(\mathcal{C})] = K_{k-1}$, and the equality case in Claim 1 holds. Next we assume that $c \geq 2$. Let $A = \{a_1, a_2, \dots, a_r\}$. We first assert that

$$\tilde{N}_X(A \setminus \{a_1\}) = \tilde{N}_X(A \setminus \{a_2\}) = \dots = \tilde{N}_X(A \setminus \{a_r\}). \quad (3.4)$$

In fact, since inequality (3.3) becomes equality, for $j = 1$, we have $|\mathcal{B}_1| = r(c - 1)$, and thus

$$|\tilde{N}_X(A \setminus \{a_i\})| = c - 1, \text{ for each } a_i \in A. \quad (3.5)$$

If $c = 2$, then (3.5) gives $|\tilde{N}_X(A \setminus \{a_i\})| = 1$ for each $a_i \in A$. Suppose to the contrary that $\tilde{N}_X(A \setminus \{a_i\}) \neq \tilde{N}_X(A \setminus \{a_j\})$ for some $i \neq j$. Then the two cliques $(A \setminus \{a_i\}) \cup \tilde{N}_X(A \setminus \{a_i\})$ and $(A \setminus \{a_j\}) \cup \tilde{N}_X(A \setminus \{a_j\})$ have intersection of order $r - 2$, which contradicts Lemma 3.1(ii). Then (3.4) holds for $c = 2$.

If $c \geq 3$, then for any two vertices $a_i, a_j \in A$, by definition, we have

$$\tilde{N}_X(A \setminus \{a_i\}) \cup \tilde{N}_X(A \setminus \{a_j\}) \subseteq \tilde{N}_X(A \setminus \{a_i, a_j\}). \quad (3.6)$$

Since $|A \setminus \{a_i, a_j\}| = r - 2 \geq t$, (3.2) means that $|N_X(A \setminus \{a_i, a_j\})| \leq c - 1$. This, together with (3.5) and (3.6), implies that

$$\tilde{N}_X(A \setminus \{a_i\}) = \tilde{N}_X(A \setminus \{a_j\}) \text{ for any two vertices } a_i, a_j \in A,$$

which completes the proof of (3.4).

For a clique $B \in V(\mathcal{C}) \setminus \{A\}$, we choose a vertex $a_k \in A \setminus B$. By definition,

$$\tilde{N}_X(A \setminus \{a_k\}) \subseteq \tilde{N}_X(A \cap B). \quad (3.7)$$

We again apply Lemma 3.1(ii) to obtain $|A \cap B| \geq t$. With (3.2) this gives $|\tilde{N}_X(A \cap B)| \leq c - 1$. This, together with (3.7) and (3.5), implies that

$$\tilde{N}_X(A \setminus \{a_k\}) = \tilde{N}_X(A \cap B), \text{ and thus } B \setminus A \subseteq \tilde{N}_X(A \cap B) = \tilde{N}_X(A \setminus \{a_k\}).$$

By the arbitrariness of B and (3.4), we have that $U(\mathcal{C}) = A \cup \tilde{N}_X(A \setminus \{a_1\})$, which, together with (3.5) yields $|U(\mathcal{C})| = r + c - 1 = k - 1$. Note that the number of r -cliques in $G[U(\mathcal{C})]$ is $\binom{k-1}{r}$. Then $G[U(\mathcal{C})] = K_{k-1}$, and the proof of Claim 1 is complete. $\square$

Now we come back to show our result. By definition, $U(\mathcal{C}_i) \cap U(\mathcal{C}_j) = \emptyset$ for distinct i, j . Thus $\sum_{i=1}^m |U(\mathcal{C}_i)| \leq n$. We write

$$I = \{i : |U(\mathcal{C}_i)| \leq k - 1\} \text{ and } J = \{i : |U(\mathcal{C}_i)| \geq k\}.$$

If $i \in I$, then trivially $|\mathcal{C}_i| \leq \binom{|U(\mathcal{C}_i)|}{r}$, and if $i \in J$, then by Claim 1, $|\mathcal{C}_i| \leq \binom{k-1}{r}$. Consequently,

$$\mathcal{N}(K_r, G) \leq \sum_{i \in I} \binom{|U(\mathcal{C}_i)|}{r} + |J| \binom{k-1}{r}. \quad (3.8)$$

Note that

$$\sum_{i \in I} |U(\mathcal{C}_i)| + |J|(k-1) \leq \sum_{i \in I} |U(\mathcal{C}_i)| + \sum_{i \in J} |U(\mathcal{C}_i)| \leq n = a(k-1) + b.$$

Hence,

$$\sum_{i \in I} |U(\mathcal{C}_i)| \leq (a - |J|)(k-1) + b. \quad (3.9)$$

Since $|U(\mathcal{C}_i)| \leq k - 1$ for each $i \in I$, by (3.8) and Lemma 2.5, it follows that

$$\mathcal{N}(K_r, G) \leq \sum_{i \in I} \binom{|U(\mathcal{C}_i)|}{r} + |J| \binom{k-1}{r} \leq (a - |J|) \binom{k-1}{r} + \binom{b}{r} + |J| \binom{k-1}{r} = a \binom{k-1}{r} + \binom{b}{r},$$

as desired.

Now we characterize the extremal graphs. Assume that G is an extremal graph. Then every inequality above becomes equality. Thus

$$\text{for each } i \in I, \text{ we have } |\mathcal{C}_i| = \binom{|U(\mathcal{C}_i)|}{r}, \quad (3.10)$$

and for each $j \in J$ (if any) we have $|\mathcal{C}_j| = \binom{k-1}{r}$. Suppose to the contrary that there exists $j \in J$. Then by Claim 1, $G[U(\mathcal{C}_j)] = K_{k-1}$, which contradicts the definition of J . This implies that $J = \emptyset$. Then from (3.10), we have

$$G[U(\mathcal{C}_i)] = K_{|U(\mathcal{C}_i)|} \text{ for each } 1 \leq i \leq m. \quad (3.11)$$

If $0 \leq b < r$, then from (3.9), $J = \emptyset$ and Lemma 2.5(i), we have that

$$m = a \quad \text{and} \quad |U(\mathcal{C}_1)| = \cdots = |U(\mathcal{C}_a)| = k - 1.$$

Since G is T -free, for each $1 \leq i \leq a$, there exists no edge with one end in $U(\mathcal{C}_i)$ and the other in $V(G) \setminus U(\mathcal{C}_i)$. This together with (3.11) implies that $G = aK_{k-1} + H$, where H is an arbitrary graph of order b , as desired.

If $r \leq b < k - 1$, then by Lemma 2.5(ii), we have that

$$\{|U(\mathcal{C}_1)|, \dots, |U(\mathcal{C}_m)|\} = \underbrace{\{k - 1, \dots, k - 1, b\}}_a.$$

This, combined with the fact that G is T -free and (3.11), implies that $G = aK_{k-1} + K_b$. The proof of Theorem 1.1 is complete. $\square$

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Declaration

Declaration of AI Use During the preparation of this manuscript, the authors used ChatGPT 5.6 to assist with language editing, grammar, clarity, and formatting. The authors reviewed and verified all AI-assisted edits and take full responsibility for the content of the manuscript.

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